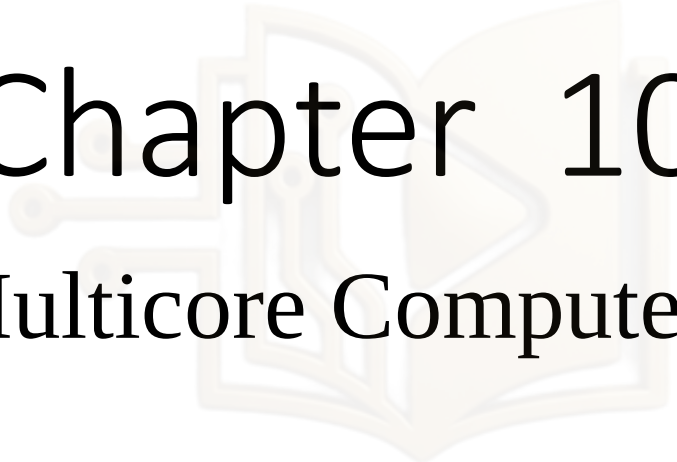




Chapter 10

Multicore Computers

+ Multi-Core Computer

11

- A multi-core processor is a processing system composed of two or more independent cores (or CPUs). The cores are typically integrated onto a single integrated circuit die (known as a chip multiprocessor or CMP).
- A many-core processor is one in which the number of cores is large enough that traditional multiprocessor techniques are no longer efficient
 - Somewhere in the range of several tens of cores - and likely requires a network on chip.

+ Multi-Core Computer

12

- dual-core processor contains two independent microprocessors.
- A dual core set-up is somewhat comparable to having multiple, separate processors installed in the same computer.
 - But because the two processors are actually plugged into the same socket, the connection between them is faster.
- Ideally, a dual core processor is nearly twice as powerful as a single core processor.
 - In practice, performance gains are about 50%:
 - A dual core processor is likely to be about one-and-a-half times as powerful as a single core processor.

+ Multi-Core Computer

1

- A multi-core processor implements multiprocessing in a single physical package.
 - Cores may or may not share caches
 - May implement message passing or shared memory inter-core communication methods.
- All cores are identical in symmetric multi-core systems.
 - EX: Intel Core 2 Duo
- They are not identical in asymmetric multi-core systems.
 - EX: IBM Cell Processor

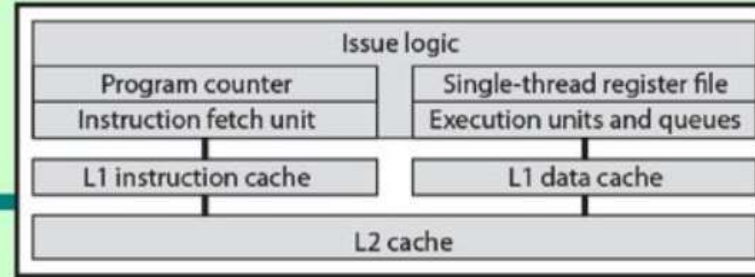
+ CMP benefits

- with a shared on-chip cache memory, communication events can be reduced to just a handful of processor cycles.
- therefore with low latencies, communication delays have a much smaller impact on overall performance.
- threads can also be much smaller and still be effective.
- automatic parallelization more feasible.

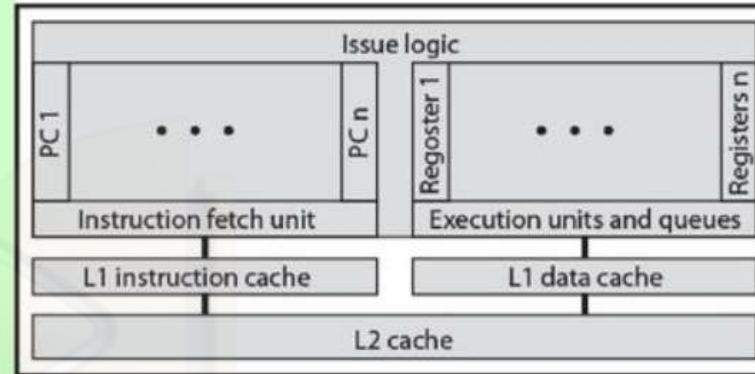
Hardware Performance Issues

- Microprocessors have seen an exponential increase in performance
 - Improved organization
 - Increased clock frequency
- Increase in Parallelism
 - Pipelining
 - Superscalar
 - Simultaneous multithreading (SMT)
- Diminishing returns
 - More complexity requires more logic
 - Increasing chip area for coordinating and signal transfer logic
 - Harder to design, make and debug

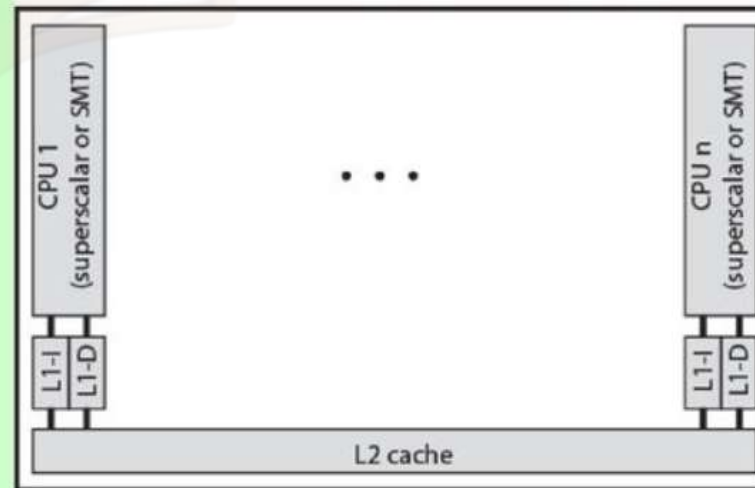
Alternative Chip Organizations



(a) Superscalar

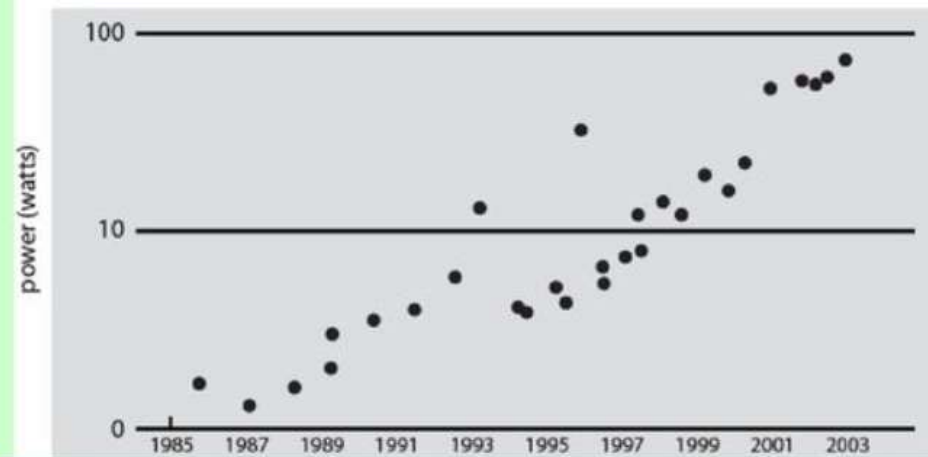
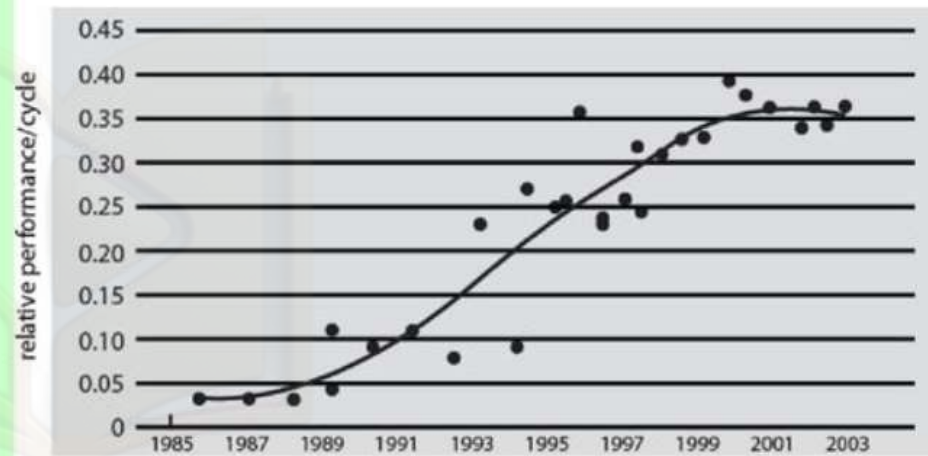
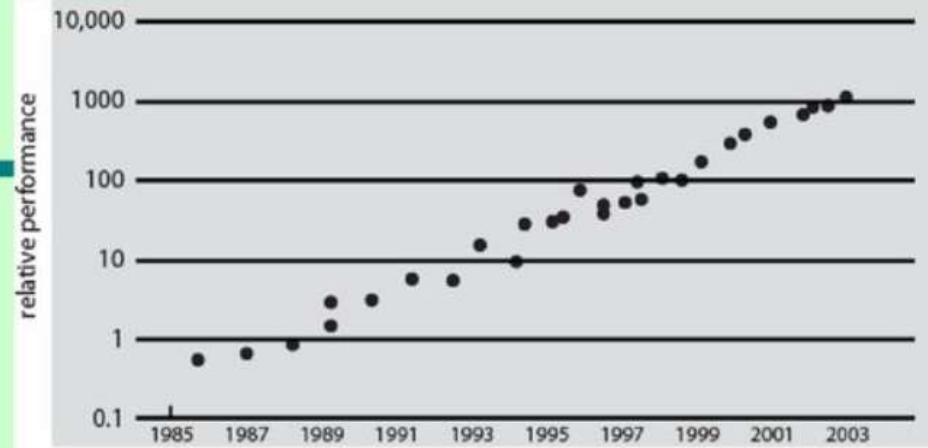


(b) Simultaneous multithreading



(c) Multicore

Intel Hardware Trends

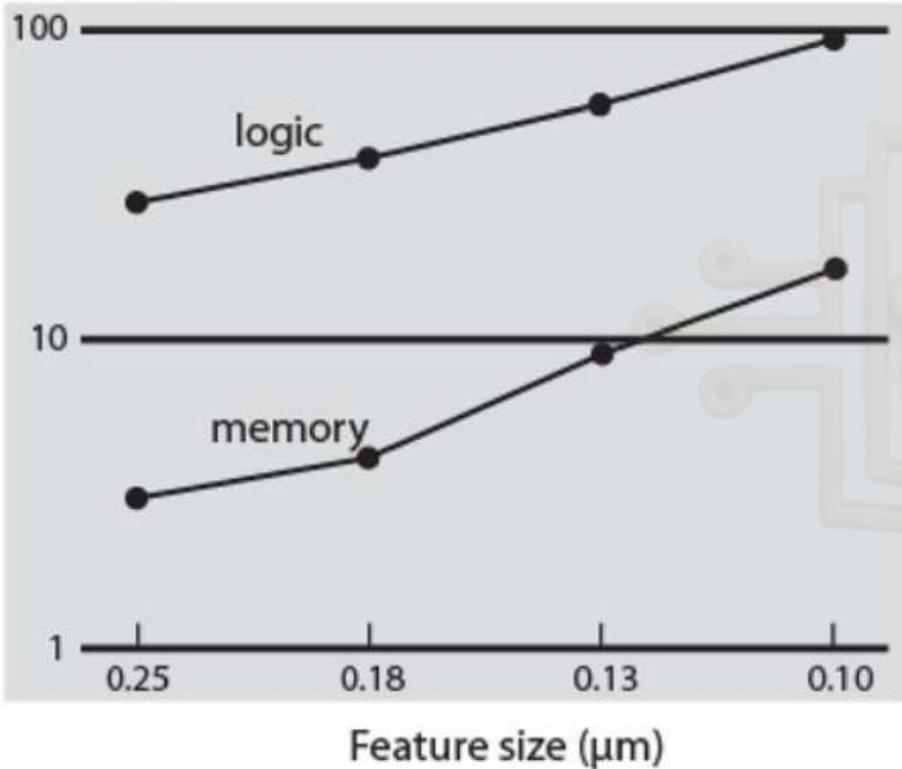


Increased Complexity

- Power requirements grow exponentially with chip density and clock frequency
 - Can use more chip area for cache
 - Smaller
 - Order of magnitude lower power requirements
- By 2015
 - 100 billion transistors on 300mm² die
 - Cache of 100MB
 - 1 billion transistors for logic
- Pollack's rule:
 - Performance is roughly proportional to square root of increase in complexity
 - Double complexity gives 40% more performance
- Multicore has potential for near-linear improvement
- Unlikely that one core can use all cache effectively

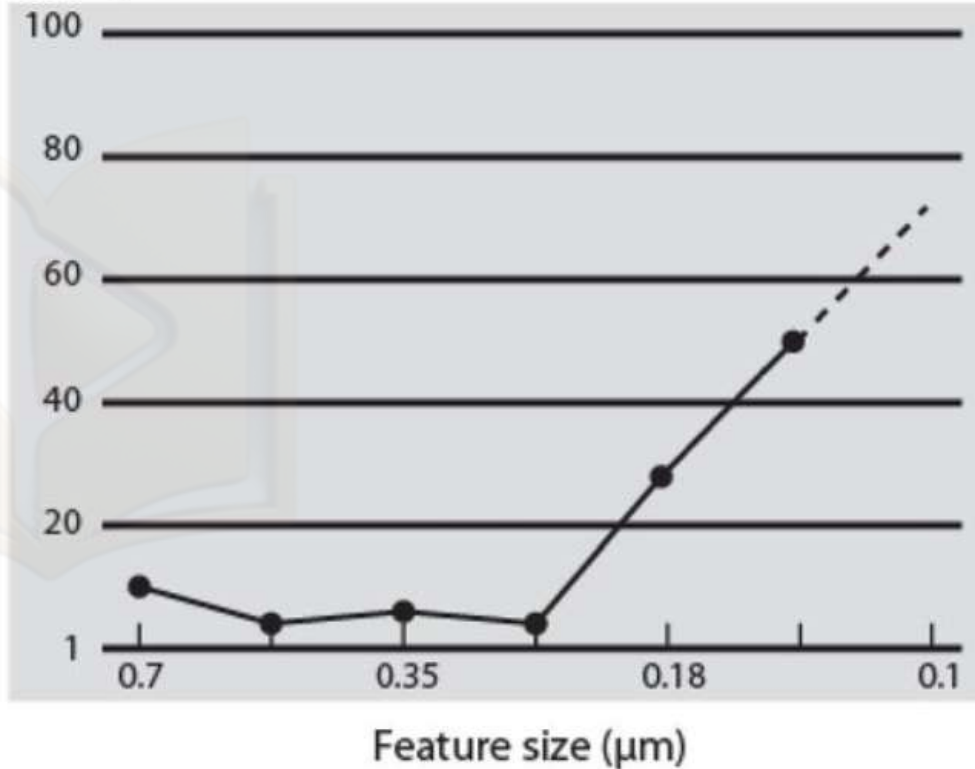
Power and Memory Considerations

Power density
(watts/cm²)



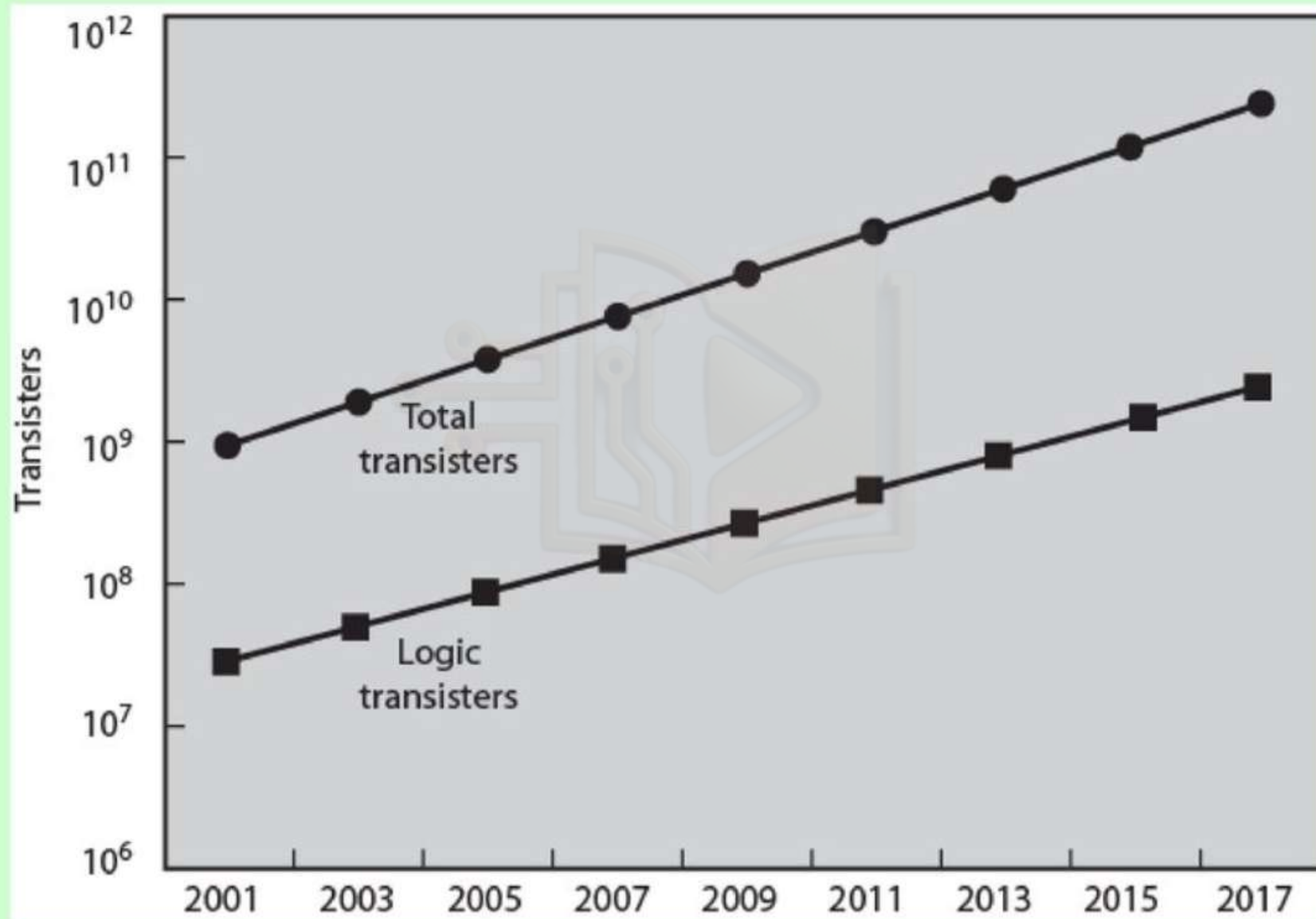
(a) Power density

cache percent
of full chip area



(b) Chip area

Chip Utilization of Transistors



Software Performance Issues

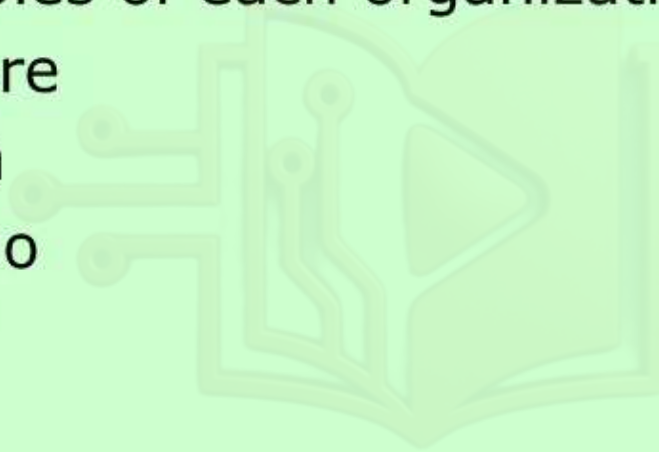
- Performance benefits dependent on effective exploitation of parallel resources
- Even small amounts of serial code impact performance
 - 10% inherently serial on 8 processor system gives only 4.7 times performance
- Communication, distribution of work and cache coherence overheads
- Some applications effectively exploit multicore processors

Effective Applications for Multicore Processors

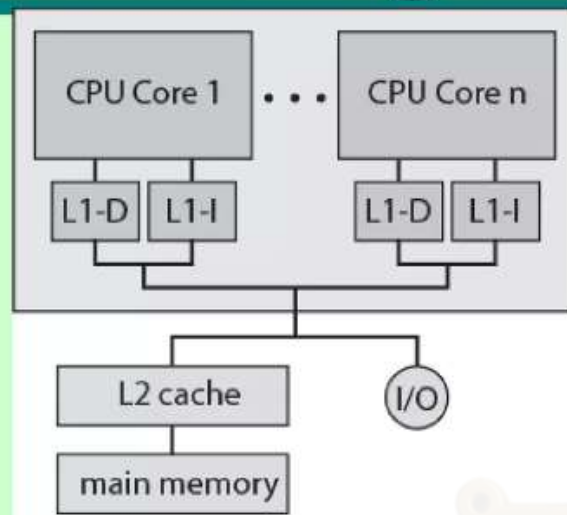
- Database
- Servers handling independent transactions
- Multi-threaded native applications
 - Lotus Domino, Siebel CRM
- Multi-process applications
 - Oracle, SAP, PeopleSoft
- Java applications
 - Java VM is multi-thread with scheduling and memory management
 - Sun's Java Application Server, BEA's Weblogic, IBM Websphere, Tomcat
- Multi-instance applications
 - One application running multiple times

Multicore Organization

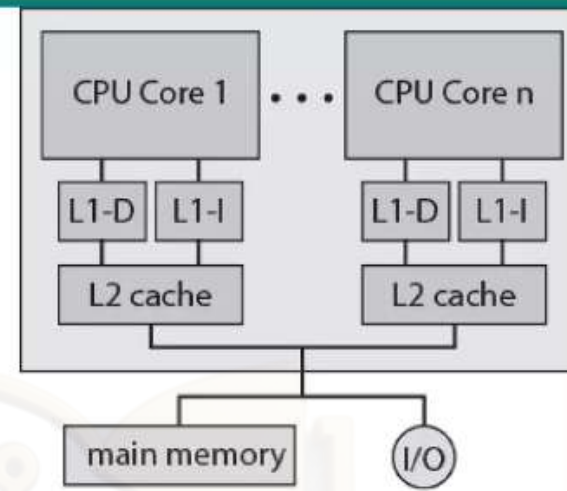
- Number of core processors on chip
- Number of levels of cache on chip
- Amount of shared cache
- Next slide examples of each organization:
- (a) ARM11 MPCore
- (b) AMD Opteron
- (c) Intel Core Duo
- (d) Intel Core i7



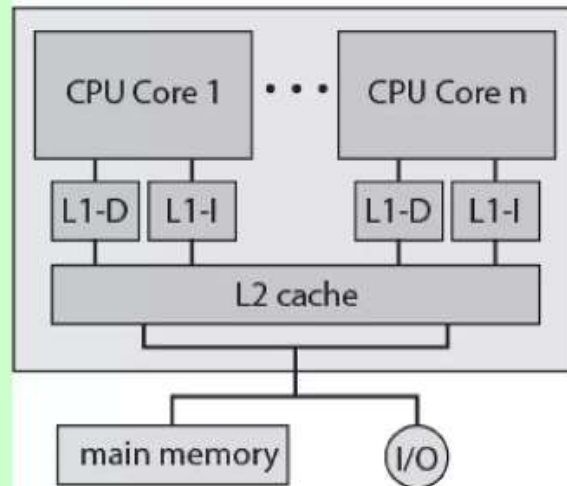
Multicore Organization Alternatives



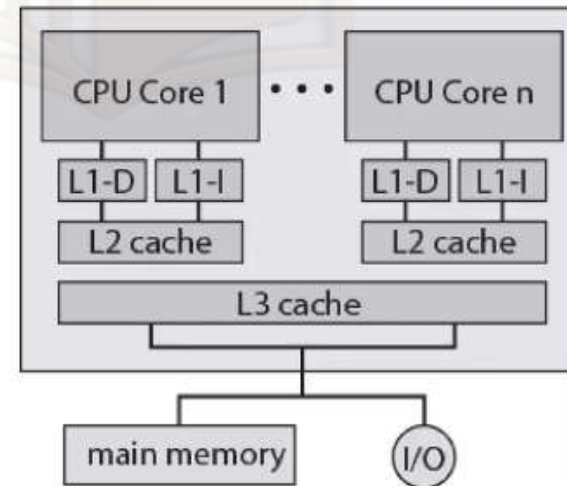
(a) Dedicated L1 cache



(b) Dedicated L2 cache



(c) Shared L2 cache



(d) Shared L3 cache

Advantages of shared L2 Cache

- Constructive interference reduces overall miss rate
- Data shared by multiple cores not replicated at cache level
- With proper frame replacement algorithms mean amount of shared cache dedicated to each core is dynamic
 - Threads with less locality can have more cache
- Easy inter-process communication through shared memory
- Cache coherency confined to L1
- Dedicated L2 cache gives each core more rapid access
 - Good for threads with strong locality
- Shared L3 cache may also improve performance